JACK J. BURLESON

AI / MACHINE LEARNING • DATA SCIENCE

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PROFESSIONAL SUMMARY

M.S. candidate in Psychology: Data Science in Human Behavior at UW–Madison (expected Dec 2026); B.S. in Psychology (neuroscience concentration, sociology minor) from UW–Parkside. I build AI systems from start to finish: multi-agent document pipelines, transformer fine-tuning and calibration, LLM evaluation, computer vision classification, and behavioral prediction models. My capstone, with industry partner American Family Insurance, looks for ways to cut the cost of AI inference and improve document processing. I also bring five years of high-volume logistics and operations work in quality assurance, documentation, and process improvement.

TECHNICAL SKILLS

Programming Languages	Python (pandas, NumPy, scikit-learn, PyTorch, Hugging Face), R, MATLAB, SQL, Parquet, Git
Machine Learning	Transformer fine-tuning, model evaluation & calibration, cross-validation, SHAP/feature importance, error analysis, computer vision (OCR-based document classification), experimental design
AI Engineering & Development	LangGraph, multi-agent pipelines, prompt engineering, LLM eval harnesses & LLM-as-a-judge, model serving (vLLM, Ollama, LM Studio, OpenRouter), ONNX + FastAPI deployment on Modal
Engineering & Cloud	Docker, pytest, Quarto, AWS, Azure

PROJECTS

Digital-Mailroom *DSHB Capstone project — American Family Insurance (industry partner)*

- Capstone with American Family Insurance to improve how documents are read, sorted, and managed: 10 Python packages covering entity extraction, classification, routing, and scalable workflows.
- Designed and built one LangGraph pipeline that joins legal-document routing, prompt experiments, deterministic scoring, dataset management, and deployment in a reproducible Docker environment.

Mailroom-ML *ML Training, Evaluation & Fine-Tuning*

- Built a ModernBERT document classifier with training, evaluation, and fine-tuning pipelines, including data lineage tracking, probability calibration, and production routing logic.
- Wrote PyTorch training pipelines with Transformers and Hugging Face datasets, evaluated with scipy, and exported to ONNX for FastAPI serving on Modal. Kept pytest coverage throughout.

RVL-CDIP-Classifier *Computer Vision & Prompt Engineering*

- Built a Python toolkit that converts TIFF/PDF pages to 300-DPI images, runs OCR, and sorts documents into 16 RVL-CDIP categories using OpenRouter vision models.
- Ran 17+ prompt iterations and produced confusion matrices, misclassification analyses, and cost projections to choose prompts and models on evidence.

Psychometric Digital Twin *Behavioral ML Research | UW–Madison*

- Built a behavioral prediction pipeline on 241 analysed survey respondents (252 matched). Ridge, Elastic Net, k-NN, Random Forest, histogram gradient boosting, XGBoost, and MLP models predicted communication-apprehension scores from demographic, employment, geographic, and transit features, with cross-validation and SHAP feature importance.
- Checked the PRCA's psychometric structure with factor analysis, then built LLM "digital twin" personas with tiered embodiment prompts (OpenRouter/vLLM). Benchmarked them against the ML baselines on shared metrics (MAE, band accuracy, F1) and analyzed systematic bias by demographic group.

EDUCATION

University of Wisconsin–Madison *Madison, WI* Expected Dec 2026

M.S. Candidate — Psychology: Data Science in Human Behavior · **GPA 3.85**

- Graduate work in machine learning, statistics, behavioral data analysis, predictive modeling, experimental design, and publication-quality visualization.

University of Wisconsin–Parkside *Kenosha, WI* Sep 2020 – May 2025

B.S., Psychology · Concentration: Neuroscience · **GPA 3.93**

- Minor in Sociology; Provost's List (7x), Dean's List (8x), Distinguished Achievement Award (2025), and Psychological Research Certification (300 lab hours).

PROFESSIONAL EXPERIENCE

UW–Parkside — Undergraduate Researcher, Cognitive Neuroscience Lab *Kenosha, WI*
Jun 2024 – Aug 2025

- Conducted cognitive neuroscience research under Dr. Melissa Gregg: designed and pilot-tested a study of perceptions of chatbot truth, deception, and user cognitive workload (code: github.com/Exios66/truth-deception-architecture).
- Ran EEG systems and analyzed neural time-series data to measure brain activity, adding signal-processing and statistical analysis to existing lab research on sleep quality and cognitive load.
- Presented findings at the 2025 Midwestern Psychological Association conference.

UW–Madison Recreation & Wellbeing — Building Services *Madison, WI* Jan 2026 – Present

- Keep recreation and wellbeing centers across campus clean, safe, and ready for use.
- Fix facility and equipment problems across multiple locations so centers stay safe and open during peak hours.
- Worked with operations and event staff to resolve facility problems quickly and limit disruption for students, staff, and visitors.

Avis Budget Group — Driver *Madison, WI* Jun 2026 – Present

- Move, stage, and safety-check rental vehicles across a 500+ vehicle fleet so cars are ready and on time across the rental network.
- Coordinate vehicle moves between locations and uphold fleet safety standards.

Meijer Distribution — Logistics Coordinator *Pleasant Prairie, WI* Jul 2020 – Dec 2025

- Promoted from Retail Loader to Logistics Coordinator for reliable quality assurance and data-driven process documentation in high-volume operations.
- Analyzed quality-assurance data for 64 daily outbound trailers across the Midwest regional network and held error rates below 1% through root-cause investigation and process refinement.
- Worked with warehouse and transportation teams to find quality problems and resolve logistics exceptions before shipment.

RESEARCH TRAINING

CITI Program: Conflicts of Interest training 2025

CITI Program: IRB Training for Students and Faculty 2025